

Integration, Reporting, and Automation

Introduction to Data Warehousing Concepts

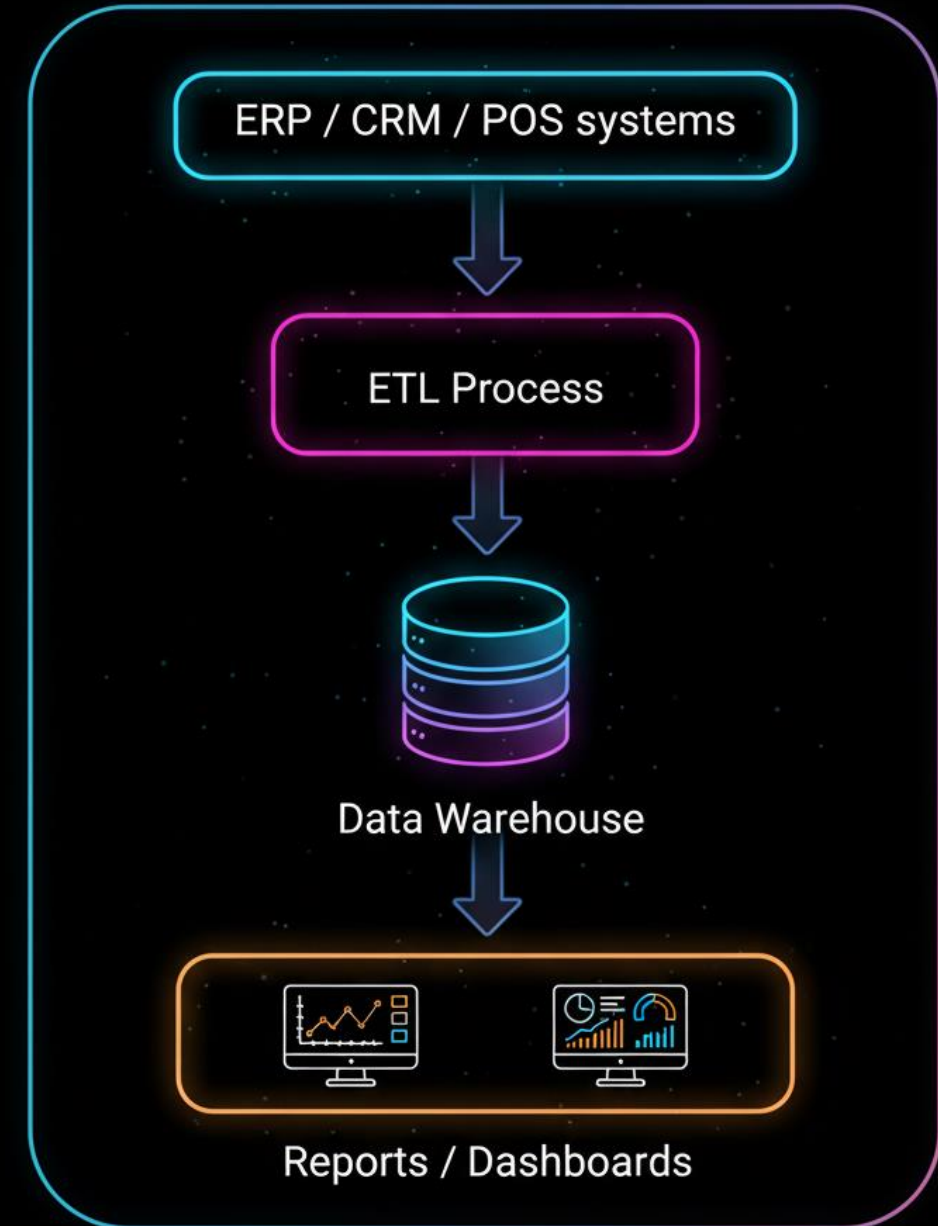


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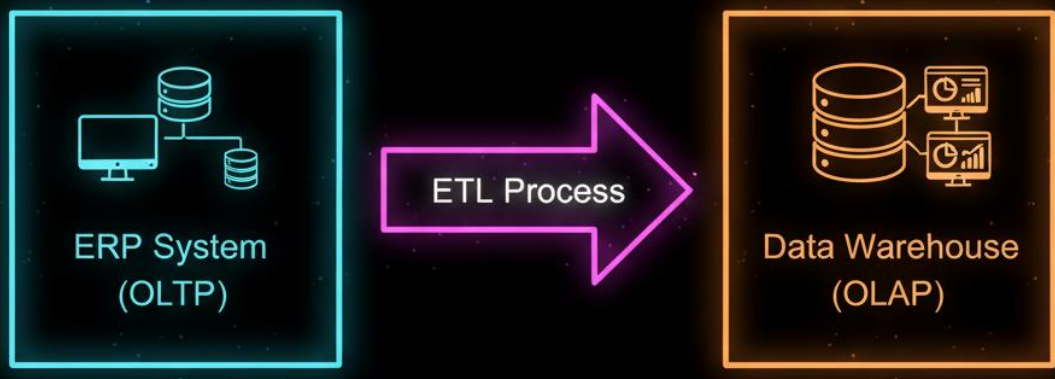
Understanding Data Warehousing

- A Data Warehouse is a centralized storage system that collects, organizes, and stores historical data from different systems
- Designed for analysis and reporting, not day-to-day operations
- Data is integrated from multiple sources (ERP, CRM, POS, etc.)
- Typically stores historical data for trend analysis.



Data Warehouse vs. Operational Database

- **OLTP**: Online Transaction Processing (operational)
- **OLAP**: Online Analytical Processing (analytical)



Feature	Operational Database (OLTP)	Data Warehouse (OLAP)
Purpose	Run daily business operations	Analyze and report on historical data
Data Type	Current, detailed, constantly changing	Historical, summarized, mostly read-only
Optimization	Fast inserts, updates, deletes	Fast read queries and aggregations
Examples	Sales order entry, inventory updates, customer payments	Monthly sales, yearly trend analysis
Typical Users	Cashiers, warehouse staff, operations team	Executives, analysts, decision-makers